The mathematics behind `insulated_monopole_Erms.m`

Time-RMS electric field in a Phytagel phantom below an insulated 2.45 GHz monopole in water

BMET3921 – microwave catheter heating project

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1 Purpose and overview

The script `V5_convection/Insulated_monopole_Erms/insulated_monopole_Erms.m` computes the time-RMS electric field E_{rms} in the vertical plane of the gel phantom that lies directly under a 2.45 GHz monopole. The antenna lies horizontally in circulating water just above the gel surface. The field map is the source term for the dielectric heating of the gel ($Q = \sigma_{\mathrm{eff}} E_{\mathrm{rms}}^2$). It was built to reproduce the photographed run (50 W generator setting, continuous wave, 6 minutes), in which the heated region in the gel peaks about 3 mm *in front of* the feed point instead of directly beneath it.

The earlier cylindrical master formula (`Dielectric_heating/cylindrical_master_formula.m`) assumed a uniform cylindrical power flow $P/(2\pi r L)$. That form has no dependence on position along the antenna, so it cannot place a hot spot anywhere along the element. The model described here replaces it with a physical current distribution. The calculation has four stages, and each is derived below:

1. the complex wavenumber k_L of the current travelling along an *insulated* conductor immersed in water, from King's insulated-antenna theory;
2. the input impedance of the element, and from it the feed current I_0 that delivers the measured power;
3. the current $I(x')$ and line charge $q(x')$ along the element and the exposed feed block;
4. the electric field these sources produce in the gel, from the retarded potentials, with a correction for the water–gel interface.

Section 3 lists every input number with its source. Assumptions are stated where they enter the derivation and collected in Section 10.

2 Conventions

All fields are time-harmonic, and the script uses the engineering convention

$$\mathbf{E}(\mathbf{r}, t) = \mathrm{Re}\left\{ \mathbf{E}(\mathbf{r})\, e^{\mathrm{j}\omega t} \right\}, \qquad \omega = 2\pi f, \quad f = 2.45 \text{ GHz}, \tag{1}$$

where the phasor $\mathbf{E}(\mathbf{r})$ holds the *peak* amplitude. With this convention a lossy medium has the complex relative permittivity

$$\varepsilon_r = \varepsilon' - \mathrm{j}\varepsilon'', \qquad \sigma_{\mathrm{eff}} = \omega \varepsilon_0 \varepsilon'', \tag{2}$$

so that all conduction and dielectric loss appear in ε'' [6]. The wavenumber in each medium is

$$k = k_0 \sqrt{\varepsilon_r}, \qquad k_0 = \frac{\omega}{c_0} = 51.35 \text{ rad/m}, \tag{3}$$

taking the root with $\operatorname{Im} k < 0$ so that an outgoing wave $e^{-\mathrm{j}kR}$ decays. For the three media this gives

$$k_{\mathrm{ins}} = 74.41 \text{ rad/m (PTFE)}, \qquad k_w = 439.5 - 26.4\mathrm{j} \text{ rad/m (water)},$$
$$k_g = 344.4 - 51.7\mathrm{j} \text{ rad/m (gel)}.$$

In the gel the wavelength is $2\pi/\operatorname{Re} k_g = 18.2\,\mathrm{mm}$ and the field $1/e$ decay length is $1/|\operatorname{Im} k_g| = 19.4\,\mathrm{mm}$. Both are larger than the 10 mm deep region that is mapped. The field is therefore neither purely quasi-static nor strongly attenuated, and the full retarded Green's function is kept in Section 7.

King and co-workers [1, 2] use the physics convention $e^{-i\omega t}$. Their formulas are converted here by the substitutions $i \to -\mathrm{j}$ and $H_n^{(1)} \to H_n^{(2)}$, where the Hankel function of the second kind represents an outgoing wave under $e^{\mathrm{j}\omega t}$.

3 Experimental constants

Table 1 lists the inputs. Values labelled "setup table" come from `V5_convection/Experimental_setup_table.xlsx`. The geometry was measured from the photograph `Gel_model_V3/mirrored.jpg` at 105.8 px/mm, because the photographed antenna is not the 27 mm antenna described in the setup table. Where the two sources differ, the table gives both values and marks the one the script uses.

The water values ($\varepsilon' = 73$, $\varepsilon'' = 8.8$) are of the size predicted by the Debye relaxation of pure water at 2.45 GHz near body temperature [9]. Treating the circulating water as pure water, with no ionic conductivity, is an assumption. If the water is saline, ε''_w would be considerably larger.

4 Stage 1: the current wavenumber on an insulated element

4.1 Why a bare-wire wavenumber is wrong

A bare conductor in water carries a current that varies along the wire roughly as $e^{\mp\mathrm{j}k_w x}$, with $k_w \approx 440\,\mathrm{rad/m}$. The corresponding wavelength is about 14 mm, so a 12 mm element would be nearly a full wavelength long. The element in the photographs, however, is the inner conductor of RG-178 with its PTFE dielectric left on. The low-permittivity sleeve changes the current's propagation constant substantially. King, Trembly and Strohbehn [1], following King and Smith [2], showed that the insulated conductor behaves as a lossy transmission line whose "outer conductor" is the surrounding medium.

4.2 Transmission-line derivation

Let the conductor of radius a be covered by insulation of permittivity $\varepsilon_{\mathrm{ins}}$ out to radius b, and be surrounded by water. The current $I(x)$ and the voltage $V(x)$ between the conductor and the ambient medium obey the telegrapher's equations [5]

$$\frac{\mathrm{d}V}{\mathrm{d}x} = -z\,I, \qquad \frac{\mathrm{d}I}{\mathrm{d}x} = -y\,V, \qquad \Rightarrow \qquad \frac{\mathrm{d}^2 I}{\mathrm{d}x^2} + k_L^2 I = 0, \quad k_L^2 = -zy, \tag{4}$$

where z and y are the series impedance and shunt admittance per unit length.

Shunt admittance. The radial displacement current passes through the insulation and then the water, which are in series. King's theory assumes [1]

$$\left| \frac{k_w}{k_{\mathrm{ins}}} \right|^2 \gg 1, \tag{5}$$

Table 1: Constants used by `insulated_monopole_Erms.m`.

Quantity	Symbol	Value used	Source / remark
Excitation			
Microwave frequency	f	2.45 GHz	Setup table
Generator power setting	P_{set}	50 W	Photographed run (the setup table's starting condition is 60 W)
Reflected power	P_{refl}	0 W	Setup table
Power dropoff (fraction lost before the antenna)	P_{dropoff}	0.7	Setup table; a tunable estimate, so 30% reaches the antenna
Delivered power	P	15 W	$P_{\text{set}}(1 - P_{\text{dropoff}})$
Media			
Gel relative permittivity, real	ε'_g	43.98	Setup table
Gel relative permittivity, imaginary	ε''_g	13.5	Setup table
Gel conductivity	σ_g	(1.7 S/m)	Setup table; not used directly. Eq. (2) gives 1.84 S/m from ε''_g
Water relative permittivity, real	ε'_w	73	Setup table
Water relative permittivity, imaginary	ε''_w	8.8	Setup table
Water temperature	T_w	37 °C	Setup table (context only)
Water flow, circulation / irrigation	–	5 / 0.03 L/min	Setup table (context only; not used in the field model)
PTFE relative permittivity	ε_{ins}	2.1	Standard value for PTFE [8]; loss neglected
Geometry			
Element length, feed to tip	L	12 mm	Photograph (the setup table's antenna is 27 mm)
Inner conductor radius	a	0.15 mm	RG-178 centre conductor, 0.30 mm diameter [10]
PTFE insulation outer radius	b	0.42 mm	RG-178 dielectric, ≈0.84 mm diameter [10]
Exposed metal feed block	l_f	1.5 mm	Photograph (solder/ferrule behind the feed)
Element axis above gel surface	h	1.1 mm	Photograph (the setup table gives 2 mm for the 27 mm antenna)
Gel block height / length	–	30 / 30 mm	Setup table; the model treats the gel as a half-space
Physical constants			
Speed of light	c_0	299 792 458 m/s	Exact (SI)
Permeability of free space	μ_0	$4\pi \times 10^{-7}$ H/m	
Permittivity of free space	ε_0	8.854×10^{-12} F/m	

Under this condition the admittance of the water path is so much larger than that of the insulation that the insulation alone sets y:

$$y = \mathrm{j}\omega C', \qquad C' = \frac{2\pi\varepsilon_0\varepsilon_{\mathrm{ins}}}{\ln(b/a)} = 113.5 \text{ pF/m}. \tag{6}$$

Here $|k_w/k_{\mathrm{ins}}|^2 = 35$, so the condition holds.

Series impedance. The series impedance has two parts. The first is the magnetic flux inside the insulation, identical to that of a coaxial line of radii a and b. The second is the impedance of the outgoing cylindrical TM wave set up in the water outside $\rho = b$. Its ratio $E_x/(2\pi b H_\phi)$ at the insulation surface is a ratio of Hankel functions [1, 2]:

$$z = \frac{\mathrm{j}\omega\mu_0}{2\pi}\left[\ln\frac{b}{a} + \frac{H_0^{(2)}(k_w b)}{k_w b\, H_1^{(2)}(k_w b)}\right]. \tag{7}$$

Wavenumber. Substituting (6) and (7) into $k_L^2 = -zy$ and using $\omega^2\mu_0\varepsilon_0\varepsilon_{\mathrm{ins}} = k_{\mathrm{ins}}^2$ gives

$$\boxed{k_L = k_{\mathrm{ins}}\left[1 + \frac{H_0^{(2)}(k_w b)}{k_w b\, H_1^{(2)}(k_w b)\,\ln(b/a)}\right]^{1/2}} \tag{8}$$

This is line 33 of the script. King's derivation also requires the insulation to be electrically thin, $|k_{\mathrm{ins}} b|^2 \ll 1$ (here $k_{\mathrm{ins}} b = 0.031$), and the outer radius to be small compared with the wavelength in the ambient medium. Here $|k_w b| = 0.185$, which satisfies the second condition.

4.3 Physical interpretation

Because $|k_w b|$ is small, the Hankel functions can be replaced by their small-argument forms [7], $H_0^{(2)}(u) \approx 1 - \frac{2\mathrm{j}}{\pi}\ln(\gamma' u/2)$ and $uH_1^{(2)}(u) \approx \frac{2\mathrm{j}}{\pi}$, with $\gamma' = e^{\gamma} \approx 1.781$. The Hankel ratio then becomes

$$\frac{H_0^{(2)}(k_w b)}{k_w b\, H_1^{(2)}(k_w b)} \approx \ln\frac{2}{\gamma' k_w b} - \mathrm{j}\frac{\pi}{2}, \tag{9}$$

and the series impedance becomes

$$z \approx \frac{\mathrm{j}\omega\mu_0}{2\pi}\ln\frac{b_{\mathrm{eff}}}{a} + \frac{\omega\mu_0}{4}, \qquad b_{\mathrm{eff}} = \frac{2}{\gamma' k_w}, \quad |b_{\mathrm{eff}}| \approx 2.6 \text{ mm}. \tag{10}$$

The first term is the inductance of a coaxial line whose outer conductor is a shell of water about 2.6 mm from the axis. The second term is a real resistance, $\omega\mu_0/4 \approx 4.8\,\mathrm{k\Omega/m}$, which accounts for the power the line leaks into the lossy ambient medium. It is this term that makes k_L complex. The approximation gives $k_L \approx 127.5 - 31.9\mathrm{j}\,\mathrm{rad/m}$, and the exact expression (8) evaluated by the script gives

$$k_L = 126.0 - 29.7\mathrm{j} \text{ rad/m}, \qquad \lambda_L = \frac{2\pi}{\mathrm{Re}\, k_L} = 49.9 \text{ mm}, \qquad \frac{\lambda_L}{4} = 12.5 \text{ mm}. \tag{11}$$

The key result is that the 12 mm insulated element is almost exactly a quarter wavelength of its own current wave. The current is therefore largest at the feed and falls to zero at the tip, and it does not oscillate along the element as a bare-wire model would predict.

5 Stage 2: input impedance and feed current

5.1 Input impedance

The characteristic impedance of the line in (4) is

$$Z_c = \sqrt{\frac{z}{y}} = \frac{\mathrm{j}k_L}{y} = \frac{k_L}{\omega C'} = 72.2 - 17.0\mathrm{j}\ \Omega. \tag{12}$$

The tip is modelled as an open circuit, $I(L) = 0$. This neglects the small fringing capacitance of the tip. A section of line of length L with an open end then has the input impedance [5]

$$Z_{\mathrm{in}} = Z_c \coth(\mathrm{j}k_L L) = -\mathrm{j}Z_c \cot(k_L L). \tag{13}$$

With $k_L L = 1.512 - 0.356\mathrm{j}$, this gives $Z_{\mathrm{in}} = 23.9 - 9.6\mathrm{j}\ \Omega$. The small reactance is consistent with a line just shorter than $\lambda_L/4$.

Equation (13) treats the monopole as a *single* open-circuited arm. In King's centre-fed insulated dipole the two arms appear in series [1]. Here there is only one arm, and the return path is assumed to close through the feed block and the surrounding water (Section 6). This is a modelling assumption. It sets the absolute scale of I_0, but it has no effect on the *shape* of the field map.

5.2 Feed current from the delivered power

The time-average power entering a port of impedance Z_{in} carrying a peak phasor current I_0 is [5]

$$P = \tfrac{1}{2}\,|I_0|^2\,\mathrm{Re}\,Z_{\mathrm{in}} \qquad \Rightarrow \qquad I_0 = \sqrt{\frac{2P}{\mathrm{Re}\,Z_{\mathrm{in}}}} = 1.12\ \mathrm{A}, \tag{14}$$

with $P = P_{\mathrm{set}}(1 - P_{\mathrm{dropoff}}) = 15\,\mathrm{W}$. I_0 is taken to be real, which fixes the phase reference. This step makes three assumptions:

- all of P is absorbed in the water and the gel, so ohmic loss in the conductor and dielectric loss in the PTFE are neglected;
- the reflected power is zero, as recorded in the setup table;
- the losses between the generator and the antenna are captured by the single factor P_{dropoff}, which is an estimate and not a measured value.

Because $E_{\mathrm{rms}} \propto I_0 \propto \sqrt{P}$, any error in P_{dropoff} rescales the whole map without moving the peak.

6 Stage 3: current and charge on the antenna

6.1 The element, $0 < x' < L$

Solving (4) with $I(0) = I_0$ and $I(L) = 0$ gives King's standing-wave current [1, 2]:

$$I(x') = I_0\,\frac{\sin\big(k_L(L - x')\big)}{\sin(k_L L)}, \qquad 0 < x' < L. \tag{15}$$

The feed is at $x' = 0$ and the tip at $x' = L$. Because k_L is complex, this sinusoid also decays slightly along the element, with a $1/e$ length of $1/|\,\mathrm{Im}\,k_L| = 34\,\mathrm{mm}$.

The charge per unit length follows from the continuity equation $\partial I/\partial x' + \mathrm{j}\omega q = 0$ [3]:

$$q(x') = \frac{\mathrm{j}}{\omega}\frac{\mathrm{d}I}{\mathrm{d}x'} = -\frac{\mathrm{j}k_L I_0}{\omega}\,\frac{\cos\big(k_L(L - x')\big)}{\sin(k_L L)}. \tag{16}$$

Because the element is nearly a quarter wave long, $\cos(k_L L)$ is small. The charge is therefore concentrated toward the tip: at the tip $|q|$ is 2.7 times its value at the feed.

6.2 The feed block, $-l_f < x' < 0$

At the feed, current leaves the inner conductor and must return along the outside of the coax. The earlier models placed this return on the braid. In V3 a 27 mm braid arm carried the full current, which made the source symmetric about the feed and put the heating peak directly under it, contrary to the photographs. The present model assumes instead that there is *no long sheath current*: the return current ramps linearly to zero across the 1.5 mm of exposed solder and ferrule behind the feed,

$$I(x') = I_0\left(1 + \frac{x'}{l_f}\right), \qquad q(x') = \frac{\mathrm{j}}{\omega}\frac{\mathrm{d}I}{\mathrm{d}x'} = \frac{\mathrm{j}I_0}{\omega l_f}, \qquad -l_f < x' < 0. \tag{17}$$

The linear profile is the simplest choice that is continuous with (15) at $x' = 0$. It is an assumption; the photographs do not resolve the actual distribution. The model is checked through total charge. Integrating q over both parts,

$$\int_{-l_f}^{L} q\,\mathrm{d}x' = \frac{\mathrm{j}}{\omega}\Big[I(L) - I(-l_f)\Big] = 0, \tag{18}$$

shows that the antenna is electrically neutral. The element carries total charge $-\mathrm{j}I_0/\omega$, and the feed block carries $+\mathrm{j}I_0/\omega$ packed into only 1.5 mm. The block's line-charge density is about 15 times that of the element next to the feed. This packed charge is the origin of the very strong field that the model predicts at the gel surface just behind the feed (Section 9).

7 Stage 4: the field in the gel

7.1 Retarded potentials

The current and charge are treated as filaments on the antenna axis (the thin-wire approximation [4]), which lies at height h above the gel surface. In a homogeneous medium of wavenumber k, the Lorenz-gauge potentials of an x-directed line source are [3, 4]

$$\mathbf{A}(\mathbf{r}) = \hat{\mathbf{x}}\,\frac{\mu_0}{4\pi}\int I(x')\,G(R)\,\mathrm{d}x', \qquad \phi(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0\varepsilon_r}\int q(x')\,G(R)\,\mathrm{d}x', \qquad G(R) = \frac{e^{-\mathrm{j}kR}}{R}, \tag{19}$$

and the electric field is

$$\mathbf{E} = -\mathrm{j}\omega\mathbf{A} - \nabla\phi. \tag{20}$$

The script evaluates the field at points (x, z) in the vertical plane that contains the antenna axis, with $z = 0$ at the gel surface and $z < 0$ inside the gel. The distance from the axis is $\rho = h - z$, and the distance to a source point is $R = \sqrt{(x - x')^2 + \rho^2}$. Because the sources are axisymmetric, the only field components are E_x (along the antenna) and E_ρ (radial, which is vertical in this plane).

The thin-wire approximation replaces the true sources, spread over radii up to $b = 0.42$ mm, by a filament on the axis. The nearest field point is at $\rho = h = 1.1$ mm, about $2.6b$. The approximation is therefore adequate in the gel but would not be valid at the insulation surface.

7.2 The gradient of the Green's function

G depends only on R, so

$$\frac{\mathrm{d}G}{\mathrm{d}R} = -\frac{(1 + \mathrm{j}kR)}{R}\,G, \qquad \frac{\partial R}{\partial x} = \frac{x - x'}{R}, \qquad \frac{\partial R}{\partial \rho} = \frac{\rho}{R}. \tag{21}$$

The $1/R^2$ part of $\mathrm{d}G/\mathrm{d}R$ is the quasi-static (Coulomb) field of the charges. The $\mathrm{j}k/R$ part is their radiating contribution.

7.3 The water–gel interface

The sources lie in water (ε_w), but the field is evaluated in the gel (ε_g). The script uses the gel Green's function, $k = k_g$ and $\varepsilon_r = \varepsilon_g$, and corrects the charge term with the quasi-static transmission factor for a planar dielectric interface. Jackson [3, §4.4] shows that for a point charge q in medium 1 near a plane boundary with medium 2, the potential in medium 2 is that of a charge

$$q'' = \frac{2\varepsilon_2}{\varepsilon_1 + \varepsilon_2}\, q \tag{22}$$

at the original position, radiating into an infinite medium 2. Setting $\varepsilon_1 = \varepsilon_w$ and $\varepsilon_2 = \varepsilon_g$, and extending the result to complex permittivities, gives

$$\tau = \frac{2\varepsilon_g}{\varepsilon_w + \varepsilon_g} = 0.768 - 0.084\mathrm{j}, \qquad |\tau| = 0.77. \tag{23}$$

The complex permittivities make this extension exactly the static result with conductivity included: the normal component of the total (conduction plus displacement) current is continuous across the boundary. The factor τ is applied to q everywhere. Charge on the insulated element also sits behind the PTFE, but outside $\rho = b$ its field is that of a line charge in water, because the normal component of $\mathbf{D}$ is continuous across the insulation surface.

No factor is applied to the current term, because μ is the same (μ_0) in water and gel, so a static magnetic image has zero strength. This choice has two consequences, both accepted as approximations:

- Scaling ϕ but not $\mathbf{A}$ means the two potentials no longer satisfy the Lorenz condition exactly. The mismatch is of the order of $1 - |\tau| \approx 0.23$ on the charge term only.
- Using k_g along the whole path R ignores the $\leq 1.1\,\mathrm{mm}$ of that path that runs through water. The difference $\mathrm{Re}(k_w - k_g) \times 1.1\,\mathrm{mm} \approx 0.10\,\mathrm{rad}$ in phase and $\mathrm{Im}(k_g - k_w) \times 1.1\,\mathrm{mm} \approx 0.03\,\mathrm{Np}$ in amplitude is small.

The model also neglects the wave reflected at the gel surface back into the water, the lateral (surface) wave of the full Sommerfeld half-space problem [11], and reflections from the bottom and sides of the 30 mm gel block. The gel is treated as a semi-infinite half-space below $z = 0$.

7.4 The field components

Substituting (19) into (20), with the interface factor applied to the charge, gives the two expressions coded on lines 64–65:

$$E_x(x,z) = -\frac{\mathrm{j}\omega\mu_0}{4\pi}\int_{-l_f}^{L} I(x')\,G(R)\,\mathrm{d}x' \;-\; \frac{\tau}{4\pi\varepsilon_0\varepsilon_g}\int_{-l_f}^{L} q(x')\,\frac{\mathrm{d}G}{\mathrm{d}R}\,\frac{x-x'}{R}\,\mathrm{d}x', \tag{24}$$

$$E_\rho(x,z) = -\frac{\tau}{4\pi\varepsilon_0\varepsilon_g}\int_{-l_f}^{L} q(x')\,\frac{\mathrm{d}G}{\mathrm{d}R}\,\frac{\rho}{R}\,\mathrm{d}x', \tag{25}$$

with $G = e^{-\mathrm{j}k_g R}/R$. The first term of (24) is the inductive field of the current. It points along the antenna and falls off slowly (roughly as $1/R$ when averaged over the element). The charge terms fall off faster ($1/R^2$ in the near zone), and their largest contribution comes from the densest charge, which is on the feed block.

7.5 Time-RMS magnitude

For a phasor field, the time average of $|\mathbf{E}(t)|^2$ over one period is $\frac{1}{2}\mathbf{E}\cdot\mathbf{E}^*$ [6], so

$$E_{\mathrm{rms}} = \sqrt{\tfrac{1}{2}\left(|E_x|^2 + |E_\rho|^2\right)}. \tag{26}$$

This is the quantity mapped and plotted. The corresponding volumetric heating in the gel is

$$Q = \sigma_{\text{eff}}\, E_{\text{rms}}^2 = \omega \varepsilon_0 \varepsilon_g''\, E_{\text{rms}}^2, \qquad \sigma_{\text{eff}} = 1.84 \text{ S/m}. \tag{27}$$

This value of σ_{eff} is about 8% above the 1.7 S/m quoted in the setup table. The difference should be resolved before the heating is calibrated in absolute terms.

8 Numerical implementation

The source integrals are evaluated with the midpoint rule on segments of $\Delta x' = 0.02$ mm: 75 segments on the feed block and 600 on the element. Each segment contributes $I(x_n')\Delta x'$ and $q(x_n')\Delta x'$, located at its centre. The field is evaluated on a 0.1 mm grid covering $-5 \le x \le L{+}5$ mm and $-10 \le z \le 0$ mm. Every field point is at least $R = h = 1.1$ mm from the axis, so the integrand is never singular. The segment length is below 2% of the smallest R, so the quadrature error is negligible compared with the modelling assumptions.

9 Results and interpretation

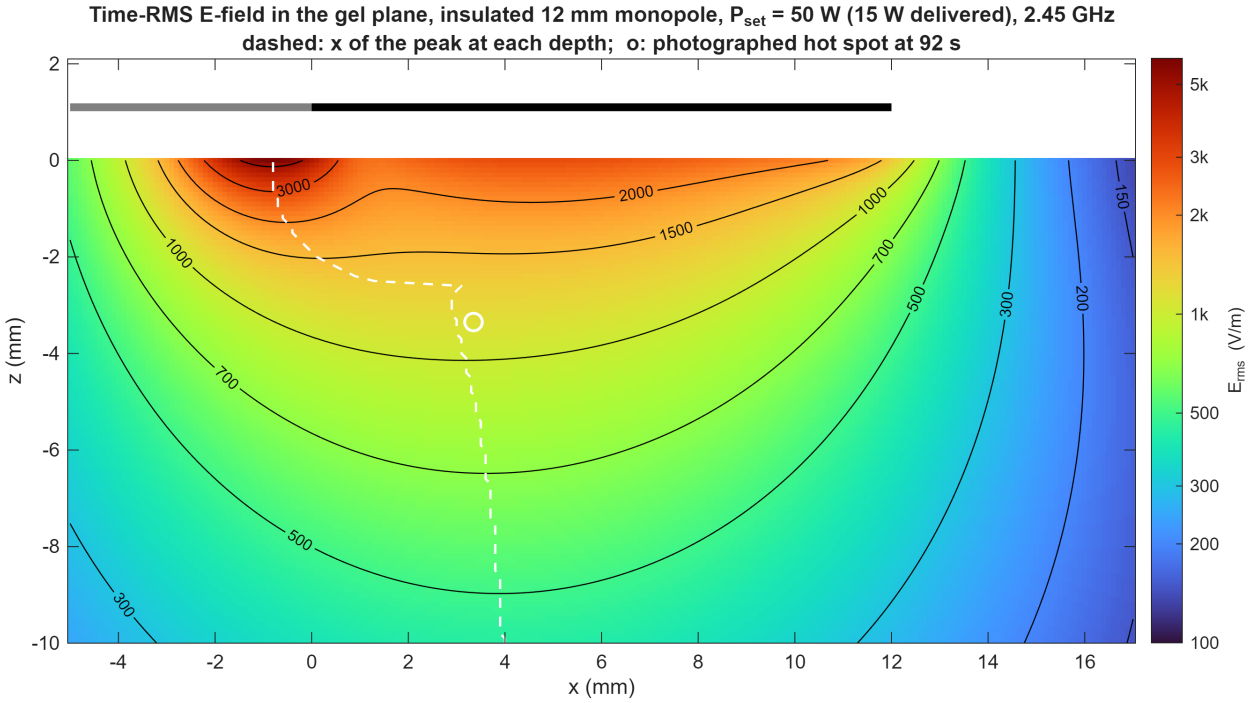


Figure 1: E_{rms} in the gel plane below the antenna, as produced by the script. Grey: feed cable. Black: 12 mm insulated element. The dashed line tracks the x position of the peak at each depth. The circle marks the hot spot in the photograph at 92 s.

The script reports $k_L = 126.0 - 29.7\text{j}$ rad/m, $Z_{\text{in}} = 23.9 - 9.6\text{j}\,\Omega$ and $I_0 = 1.12$ A. The field map (Fig. 1) has two regimes.

At the gel surface the strongest field, 5.9 kV/m, sits at $x = -0.8$ mm, directly under the feed block. Splitting (24)–(25) into their terms shows that this peak is almost entirely the vertical charge field E_ρ from the packed block charge of (17): about 5.6 kV/m, against 2.0 kV/m from the current term. At 1 mm depth the peak is still behind the feed (2.3 kV/m at $x = -0.7$ mm). By 2 mm it has moved to the feed (1.5 kV/m at $x = 0.1$ mm). The field of this charge concentration decays quickly with depth. In the experiment it also sits in the top ∼2 mm of gel, where the circulating water holds the surface near the water temperature. That is consistent with the absence of a visible hot spot there in the photographs.

Below about 2.5 mm the peak jumps forward and stays 3–4 mm in front of the feed: 1.2 kV/m at $x = 2.9$ mm and 3 mm depth, 0.87 kV/m at $x = 3.4$ mm and 5 mm depth, and 0.44 kV/m at $x = 4.0$ mm and 10 mm depth. In this regime the inductive current term dominates. At 3 mm depth it is about 1.5 kV/m, and the charge term (about 0.45 kV/m) partly cancels it. The forward shift is a geometric effect of the current distribution. A point at depth D averages the current over a window of the element roughly D wide, because the kernel $G(R)$ weights nearby sources most. Under the feed, half of that window lies behind the feed, where only the short block carries current, so the average is low. Moving forward fills the window, but the quarter-wave current (15) falls toward the tip. The product of these two effects peaks a few millimetres in front of the feed.

The photographed hot spot at 92 s sits at $(x, z) \approx (3.35, -3.35)$ mm, on the dashed peak track in Fig. 1. The photographs show the hot spot 2–4 mm in front of the feed at 2.5–4.5 mm depth over the run. A separate three-dimensional conduction calculation in this project, driven by this field with the gel surface held at the water temperature, placed the thermal peak at $(3.0, 2.25)$, $(3.25, 3.0)$ and $(3.5, 4.25)$ mm (x, depth) at 29, 92 and 332 s, in agreement with the photographs.

10 Summary of assumptions

The assumptions introduced above are collected here in the order they enter the calculation.

The inputs are taken from two sources. The geometry comes from the photograph and the RG-178 datasheet, not from the setup table, because the photographed antenna is 12 mm long with the PTFE left on. The water is treated as pure (non-saline). PTFE loss is neglected. The delivered power is the generator setting times the estimated factor $1 - P_{\mathrm{dropoff}}$, and no power is reflected.

The current wavenumber relies on King's conditions $|k_w/k_{\mathrm{ins}}|^2 \gg 1$, $|k_{\mathrm{ins}}b|^2 \ll 1$ and $|k_w b| < 1$, all of which are satisfied. The antenna is modelled as a single open-circuited arm of insulated line, with the fringing capacitance at the tip neglected. The return current is confined to the 1.5 mm feed block with a linear profile, and there is no sheath current on the braid. This last choice has the greatest influence on where the peak falls.

The field calculation uses thin-wire filament sources. It treats the gel as a homogeneous half-space and uses the gel Green's function throughout. The water–gel interface enters only through the quasi-static transmission factor τ, applied to the charge term. Reflected and lateral waves at the interface, the finite size of the gel block, and the slight violation of the Lorenz condition that τ introduces are all neglected.

Finally, the conversion to heating uses $\sigma_{\mathrm{eff}} = \omega\varepsilon_0\varepsilon_g''$. This value differs from the 1.7 S/m conductivity quoted in the setup table.

References

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